

Sources of Error for Fading of Madder Laker Pigment

The colourimetry and survey data collected for this repository has sources of error due to lack of information, human errors, and biases in the interviews.

During the production of the samples and measuring of the three sections the students had no prior knowledge of the assignment. Due to the lack of information of the intended use for the sample, some of the students painted the samples unevenly, which resulted in visible paint strokes and patchiness of the colour. This affected the survey as the first question aimed to observe whether the participants would notice the three areas without prior knowledge of the experiment, as the patchiness created confusion and several participants could not separate the patchiness with the three areas. Several of the students mentioned having to clarify the three areas through the rest of the questions, which made the questioning fairly biased. The questioning would also be less biased if executed in the same lighting for each interview with stricter guidelines for the questioning.

Students were not given information of what the colourimetry data would be used for and were instructed to measure each area five times. Since the paint layers were patchy and uneven the five measurements of the covered, uncovered and TruVue could be inaccurate, as some students did not measure the entire area, and the placement of the spectrophotometer was random and imprecise. To improve the representation of the colour the amount of measurements could have been increased and the placement could be distributed more evenly over the entire sections.

Some improvements could be done to the survey, as the questions and strategy of the interviews should have been arranged to reduce bias and make the data more comparable. The three areas were lined up, which made the first and the third section not as comparable as the others. This could affect the answers as the colour difference could be less noticeable. Instead of having them lined up, the students could have had three different swabs or cut them in three parts, so the participants could juxtapose them. This would make the first question insufficient, but this question was inadequate anyways as described previously.

The students also had individual issues, making the data inadequate. All students except one painted with the same paint and brush, making one of the swabs different from the rest. The colourimetry data differs from the rest but has been included in the complete dataset and analysis. Student 10 also cut their sample into three parts, dividing the marked areas. Their survey differs from the rest, which makes the results biased.